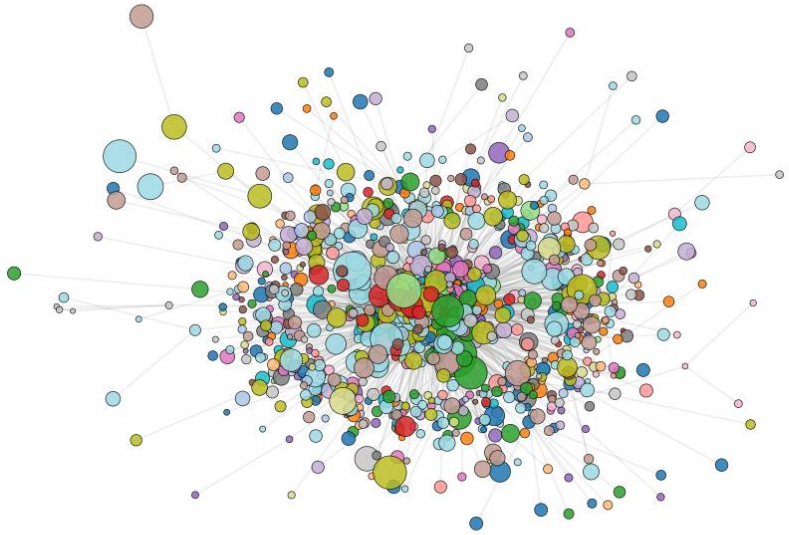
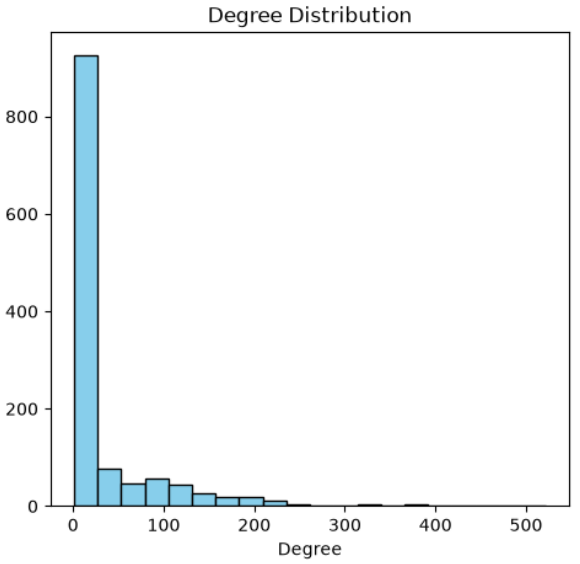
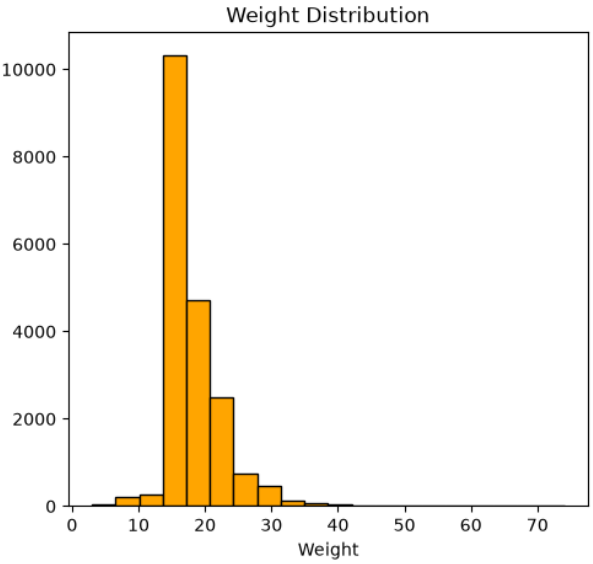
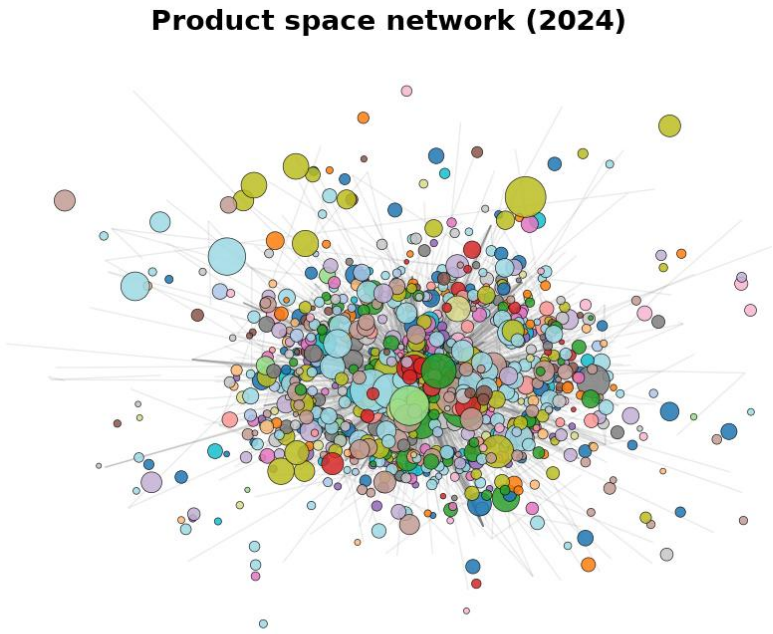
Product space network (2014)



The netwrok represents the Maximum Spanning Tree of the 2014 product space including links with threshold >= 15. Node size corresponds to export volume (square root scale). Node color refers to the product category. Source: Atlas of Economic Complexity







The netwrok represents the Maximum Spanning Tree of the 2024 product space including links with threshold >= 15. Node size corresponds to export volume (square root scale). Node color refers to the product category. Source: Atlas of Economic Complexity

The degree distribution of the network is right-skewed, typical scale-free network structure. Majority of nodes (products) have a low degree, while few nodes have an extremely high degree, acting as hubs.

The distribution of edge weights peaks around value between 15 and 20. The majority of the retained, filtered edges represent stable, medium-intensity economic connections between products (product categories).

The analysis identified a total of 21 different communities, but the structure is dominated by 3 exceptionally large communities (with 444 products, 382 products and 358 products), while the remaining communities consist of smaller, specific product groups.

The core level of the network is 72, based on which 177 products belong to the core – core products -, while the remaining products (1066) are located in the periphery – periphery products. This shows that a massive, sparser periphery is built around a dense, complex core of the product space.

Comparing the 2014 and 2024 network

Out of the 19 527 edges of the 2014's network, 10 216 edges persisted in the 2024 network as well. This represents a persistence ratio of 52,32%, which shows that around half of the product-product connections remained stable over a decade.

The correlation of the weights of the common edges is 0.6919 (with ~0 p-value) which indicates a strong, significant positive relationship. This means that edges that were strong in 2014 were likely to remain strong in 2024, therefore the intensity of the connections is fairly stable over time.

Both the 2014 and 2024 networks are characterized by a dense, solid core in the center surrounded by a sparse periphery. The overall visual density and spatial layout remains consistent across the decade, which indicates a stable macro-structure in the global product space.

The distribution of different product categories (indicated by colors) appears evenly mixed throughout the core in both years. This suggets that relatedness spans across diverse product groups rather than forming isolated single-category parts, and it shows that the underlying economic diversification patterns persisted from 2014 to 2024.

Furthermore, the 2024 network appears slightly denser and more tightly woven compared to 2014, reflecting an increase in economic connections and product linkages over the decade.